

We didn't have any conviction that the market was going higher. We almost always want to have some long beta exposure, however, and by making the option conditional on the XLP outperforming the S&P, we were able to get beta exposure to the market extremely cheaply. When you own options, you're always fighting against the time decay. Figuring out how to make the option premium cheaper is one way of mitigating that decay.

So the basic premise is that beta is measured based on daily relative price changes, which can be a very poor indicator of long-term relative price changes.

Right, a fact that is obvious if you look at a long-term chart comparison of the XLP versus the S&P. Volatility is a terrible proxy for measuring potential price change over longer intervals of time. For example, if an asset price changes by a constant percentage each day, its volatility will be zero. One of our strategies is called *cheap sigma* and is predicated on the idea that markets sometimes trend and that volatility will dramatically understate the potential price move of markets that trend. For example, in 2007, Charlie noticed that the Canadian dollar was trending very smoothly as it broke the dollar mark for the first time in decades. Spot went from about 1.10 [Canadian dollars per U.S. dollar] to about 0.92—a very large price move. The market volatility, however, was very low. Based on the volatility, a nonsensically improbable event had just occurred.

That's past tense. How does that relate to a trade you did?

If the three-month implied volatility says that the price move that has just occurred was a three and a half standard deviation event, we are going to like the odds of buying deep out-of-the-money options for a price move back in the opposite direction.

So the basic concept is that option prices will tend to be priced too low in smoothly trending markets.

Yes, and this is another type of option mispricing. The broader principle is that the explicit and implicit assumptions that go into option pricing models often diverge from the underlying reality. Looking for those divergences can be a very profitable exercise because you can wait and